

Question:

A 34-year-old woman presents to her OB/GYN with complaints of missing her last 3 periods as well as intermittent spontaneous milky-white nipple discharge bilaterally for the past 3 months. Vital signs are stable and within normal limits. Neurologic examination is without abnormality, including normal visual fields. Serology and MRI of the brain are ordered, with results pending. Which of the following sets of laboratory results would be expected in this patient?

Answer Choices:

- A. Decreased prolactin, decreased FSH, decreased LH
- B. Decreased prolactin, increased FSH, increased LH
- C. Increased prolactin, decreased FSH, increased LH
- D. Increased prolactin, decreased FSH, decreased LH

Reasoning + Answer:

<think>

Finding Reasoning Paths :

- 1. **Breast discharge (galactorrhea) + missing periods → Prolactin-secreting tumor (e.g., prolactinoma) → Hyperprolactinemia → Decreased gonadotropins. **
- 2. **Breast discharge (galactorrhea) + missing periods → Hypothalamic or pituitary issue → Disruption of HPG axis → Decreased gonadotropins. **
- 3. **Breast discharge (galactorrhea) + missing periods → Elevated prolactin levels → Decreased gonadotropin levels. **
- 4. **Breast discharge (galactorrhea) + missing periods → Prolactinoma → Elevation of prolactin → Decreased gonadotropin secretion. **

Reasoning Process :

- 1. **Clinical Presentation : **
 - The patient presents with missing periods (amenorrhea) and spontaneous galactorrhea (milky-white nipple discharge) for the past 3 months .
 - These symptoms suggest a disorder of the hypothalamic-pituitary-gonadal (HPG) axis , likely due to hyperprolactinemia .
- 2. **Pathophysiology of Galactorrhea : **
 - Galactorrhea can occur due to elevated prolactin levels , typically caused by a prolactin-secreting tumor (e.g., prolactinoma), hyperpituitarism , or exogenous factors (e.g., drugs or pregnancy).
- 3. **Pathophysiology of Amenorrhea : **
 - Amenorrhea in women of reproductive age is commonly caused by hormonal imbalances , including hyperprolactinemia .
 - Elevated prolactin levels can suppress follicle-stimulating hormone (FSH) and luteinizing hormone (LH), leading to anovulation and amenorrhea .
- 4. **Impact of Elevated Prolactin on Gonadotropins : **
 - Hyperprolactinemia is a well-known cause of hypogonadotropic hypogonadism .
 - In this condition , elevated prolactin levels directly suppress FSH and LH secretion from the pituitary gland .
- 5. **Order of Effects : **
 - Elevated prolactin → Decreased gonadotropins (FSH and LH).
- 6. **Exclusion of Other Causes : **
 - The patient has no neurological symptoms (e.g., headache , visual field defects), ruling out intracranial masses or central nervous system lesions .
 - There is no mention of drug use , stress , or other triggers of galactorrhea , making a secondary cause of hyperprolactinemia unlikely .

Conclusion :

The patient is likely experiencing **hyperprolactinemia** due to a prolactin-secreting tumor (e.g., prolactinoma). This condition leads to **decreased gonadotropin secretion**, characterized by **decreased gonadotropin levels** (FSH and LH).

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<answer>

Decreased prolactin , decreased FSH , increased LH

</answer>